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To what extent do politeness expectations shape risk perception? Even numerical probabilities are under their spell!

Miroslav Sirota [a,*], Marie Juanchich [b]

[a] School of Medicine, King's College London, Capital House, 7th floor, 42 Weston Street, SE1 3QD, London, United Kingdom
[b] Kingston Business School, Kingston University London, Kingston Hill, KT2 7LB, United Kingdom

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ABSTRACT

Politeness theory posits that uncertainty quantifiers can be interpreted as hedging strategies and thus be perceived as communicating greater certainty than when they are interpreted as likelihood-communication devices. This has only been tested with verbal probabilities (e.g., it is possible). The present paper aims to test whether numerical probabilities can also be interpreted as face-management devices and to investigate the effect of such an interpretation on risk perception. Four experiments focused on the effect of interpretations of numerical probabilities in negative outcome predictions on risk perception (e.g., there is a 50% probability that your stocks will lose their value). Politeness expectation was manipulated by the personality of the speaker (i.e., blunt vs. tactful, Experiments 1 and 2) and according to the conversational partners' need for politeness (Experiments 3 and 4). Results show that numerical probabilities, like verbal ones, were interpreted as likelihood-communication or face-management devices and that the two interpretations led to different risk perceptions. Findings were replicated with different formats, such as percentage (e.g., 50%) and chance ratio (e.g., 1 chance in 3) and with different degrees of certainty (e.g., .30, .50 and .70). Theoretical and practical implications relevant to risk communication are presented and discussed.

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1. Introduction

Traditionally, verbal probabilities (e.g., it is possible, there is a small probability) have been considered as vague likelihood-communication devices. Yet, in agreement with politeness theory (Brown & Levinson, 1987), evidence suggests that verbal probabilities are often perceived as face-management devices, providing optimistic and vague bad news to a conversational partner instead of a plain and accurate forecast (Bonnefon, Feeney, & De Neys, 2011; Bonnefon & Villejoubert, 2006; Juanchich & Sirota, submitted for publication; Juanchich, Sirota, & Butler, 2012; Pighin & Bonnefon, 2011). The present paper aims to examine whether the pragmatic use of verbal probabilities as face-management devices is limited to linguistic risk material or can be extended to numerical probabilities.

1.1. Politeness theory

Politeness theory posits that all humans need to protect and promote both a positive sense of self and freedom of action (Brown & Levinson, 1987). Moreover, the theory suggests that communication acts damaging these needs are felt to be a threat that can be diminished by the use of *face-management* strategies. For example, asking for a service may require such a strategy because the request poses a threat to the service provider's future freedom of action. Giving bad news is considered a threat for hearers because it damages their emotional integrity (e.g., creating distress). Introducing uncertainty is a face-management strategy. For example, a criticism can be signified by tactfully saying "*I am not sure* I follow you", instead of by blatantly saying "You are not clear".

1.2. Face-management and risk quantification

A number of findings show that verbal probabilities can be perceived and used as face-management devices through providing a risk assessment adjusted towards the hearer's preferences. For example, when asked to give bad news, speakers tactfully preferred to say "It is evenly probable that your stock will lose its value" whereas they knew this outcome was likely (Juanchich & Sirota, submitted for publication). Used as an optimistic face-management device, verbal probabilities therefore aim to both communicate a degree of certainty and to soften threat. Risk communicated can be adjusted upwards or downwards as a function of the valence of the outcome. Indeed, in negative outcome predictions, tactful speakers downwardly adjust the risk level, whereas, in positive predictions, speakers are expected

* Corresponding author at: Medical Decision Making and Informatics Research Group, School of Medicine, King's College London, Capital House, 7th floor, 42 Weston Street, SE1 3QD, London, United Kingdom. Tel.: +44 20 7848 6652; fax: +44 20 7848 6620.
E-mail address: miroslav.sirota@kcl.ac.uk (M. Sirota).

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to adjust the probability they communicate upwards so as to improve the hearer's prospects (Juanchich et al., 2012).

In addition to optimistic situation assessment, politeness theory suggests that tactful speakers can pretend to be uncertain to enable hearers to save face (Bonnefon & Villejoubert, 2006, Brown & Levinson, 1987). For example, tactful speakers would prefer to hedge "[I am sure] you made a mistake here" by saying "maybe you made a mistake here". Hedging appears as an extreme case of optimistic assessment, where speakers provide an adjusted degree of certainty in which the starting point is certainty. Hedging therefore implies that the face-management function of verbal probabilities could be completely separated from their likelihood-communication function.

Erev and Cohen (1990) suggest similar considerations about verbal probabilities with their "cover hypothesis". The hypothesis posits that people use verbal probabilities to "cover" themselves and avoid taking the responsibility of an accurate (numerical) estimate (e.g., saying "it is possible" instead of "there is 1 chance in 2"). The optimistic use of verbal probabilities and the cover hypothesis differ in that the optimistic assessment strategy is used to protect the recipient (from a ruthless truth) whereas the cover hypothesis postulates a preference for verbal formatting fuelled by self-serving considerations. Moreover, in contrast to hedging or optimistic assessment, the cover use of verbal probabilities does not imply the communication of a different probability, but simply the translation of an accurate belief into a vague format.

1.3. Effect of face-management interpretations

When receiving a prediction (e.g., it is possible that your stocks will lose their value), it is not uncommon for speakers to interpret the verbal probability as a tool intended to soften the threat of bad news (i.e., either hedging or optimistic) instead of as a simple likelihood-communication device. In a negative outcome risk communication, hearers who interpret verbal probabilities as face-management devices perceive a greater risk that the negative outcome will occur. For example, a patient who is told that "it is possible that you will be deaf" may either understand that the general practitioner is conveying a 50% probability or that she is tactfully trying to make a more likely prognosis of deafness (Bonnefon & Villejoubert, 2006; Juanchich et al., 2012). The interpretation of the risk quantifiers' function is not only influencing risk perception but also the severity of the situation and the decision-making of the recipient. For example, when a speaker who said "Possibly, the stocks you bought will lose their value" was believed to be using a face-management strategy, people perceived financial loss to be more likely, the situation to be more severe (Experiment 3), and were then more likely to sell the stocks in question (Experiment 4; Juanchich et al., 2012).

1.4. The danger of politeness risk miscommunication

As described above, in order to comply with social needs and satisfy politeness expectations, people do not directly express their degree of certainty and hearers apply a risk perception adjustment based on the supposed intention of speakers (Bonnefon & Villejoubert, 2006; Juanchich & Sirota, submitted for publication; Juanchich et al., 2012). Therefore, risk communication can give rise to many miscommunication issues. The danger of miscommunication lies in wrongful interpretation of the speaker's goal, and thus in the difference between risk perceived from a likelihood-communication interpretation and from a face-management interpretation (Bonnefon, Feeney & De Neys, 2011; Juanchich & Sirota, submitted for publication; Juanchich et al., 2012). For example, a doctor could announce to two patients "it is possible that you have cancer" and one patient might leave thinking of the possibility of cancer whereas the other will leave with the certainty of cancer. The danger of risk misperception is made especially salient by the lack of consensus on the interpretation to give to a verbal forecast which focuses on a negative outcome. For example, when told "it is possible that you will be deaf", around 40% of people believed that the speaker gave a plain uncertainty, whereas 60% believed that the speaker aimed to tactfully communicate a likely prognosis (Bonnefon & Villejoubert, 2006). Results of Juanchich et al. (2012) showed a similarly low level of agreement. Finally, Juanchich and Sirota (submitted for publication) showed that a similar low level of consistency applies to the speakers. They found that when asked to describe a possible investment loss, around half of the participants used verbal probabilities to plainly communicate likelihood whereas the other half had a face-management objective.

So far, the discrepancy of risk perception created by face-management and likelihood-communication interpretations has been assumed to be triggered by linguistic expressions only. Indeed, investigations conducted to broaden our understanding of the process has focused only on linguistic materials, such as verbal probabilities or general quantifiers like *some*, *all* and *few* (Bonnefon, Feeney, & Villejoubert, 2009). This assumption is implicitly supported by the fact that Brown and Levinson (1987) did not use numbers to illustrate hedging or optimism strategies, but rather linguistic expressions in the form of adverbs (e.g., maybe) or verbs (e.g., I think). The focus on linguistic material often concurs with a belief that numerical probability is a more suitable format for risk communication, because it is supposed to be more accurate and to leave less room for subjective interpretation (e.g., Bonnefon & Villejoubert, 2006). This view is supported by findings indicating that verbal probabilities have more "pragmatic power" than numbers. For example, whereas verbal probabilities direct attention unambiguously either toward the occurrence or the non-occurrence of an outcome (e.g., there is a chance vs. it is not certain), numerical probabilities are not interpreted with such a consensus (Teigen & Brun, 1995; Teigen & Brun, 2000). Windschitl and Wells (1996) even formally hypothesized that verbal probabilities elicit an intuitive mode of information processing, making them more sensitive to contextual influence, whereas numerical probabilities elicit a more rule based approach, reducing, or even removing, sensitivity to contextual influence. For example, the authors showed that people were sensitive to the number of possible alternative outcomes when providing their responses on a verbal probability scale but not when providing their responses on a numerical probability scale.

1.5. Are numerical probabilities used as face-management devices?

So far, there is very little evidence that numerical risk quantifiers do not perform face-management functions. On the contrary, previous research has shown that numerical probabilities could elicit or be applied to a range of likelihoods as a function of contextual factors such as base rate, representativeness or severity (Windschitl & Weber, 1999). Numerical probabilities are, for example, sensitive to the representativeness of an outcome. As a result, a 30% chance of snow in December conveys a higher degree of certainty than a 30% chance of snow in November (Windschitl & Weber, 1999). Moreover, Pighin, Bonnefon, and Savadori (2011) showed in a recent study that outcome severity affected the subjective interpretation of a numerical probability. In their study, pregnant women perceived that a fictitious pregnant woman had a greater risk of having her child affected by Down syndrome than by insomnia, although these two medical conditions were both described as having a 1 in 28 chance of occurring. It is worth noting that although findings suggest that numbers are sensitive to contextual factors, studies comparing the context effect on both verbal and numerical probabilities have shown that numbers are less sensitive to contextual factors than verbal probabilities (Piercey, 2009). For example, when participants were encouraged to make an optimistic risk assessment in an accounting context, they provided more optimistic assessments with verbal probabilities than with numerical ones (Piercey, 2009).

Based on the body of evidence concerning the contextual sensitivity of numerical probabilities, we propose that numerical probabilities can be influenced by politeness expectations, yet less so than verbal probabilities. Specifically, numerical probabilities qualifying negative outcomes

and interpreted as face-management devices will convey a higher probability than those interpreted as likelihood-communication tools.

1.6. Methodology of investigating politeness concerns

In most of the cases, politeness concerns have been manipulated by changing the degree of severity of the uncertain outcome. For instance, a deafness prognosis is more severe than a prognosis of insomnia and was thus expected to trigger a greater need for politeness (Bonnefon & Villejoubert, 2006; Juanchich et al., 2012; see also Pighin & Bonnefon, 2011). The severity manipulation did indeed trigger greater rates of face-management interpretations but it represents a methodological weakness because it confounds politeness concerns with other factors hypothesized to influence risk perception, such as base rate, worry or arousal. Severe outcomes are indeed perceived as more frequent and eliciting greater worry and arousal than neutral ones which all have been shown to impact risk perception (respectively, Wallsten, Fillenbaum & Cox, 1986; Weber & Hilton, 1990; Vosgerau, 2010). Furthermore, according to the asymmetric loss function account, severity has an indirect effect on risk perception through the asymmetric loss function (Harris & Corner, 2011; Harris, Corner, & Hahn, 2009; Weber, 1994). This account posits that in severe risk assessment situations, people err on the safe side by inflating their risk perceptions.

Base rate, worry, arousal, loss aversion and politeness are several factors expected to influence the effect of severity on risk perception. The confounding of their effects undermines the solidity of findings based on the severity manipulation method. To bring more solid evidence of the effect of politeness concerns, the studies presented here rely on a different, more direct, manipulation of politeness concerns.

The present research aims to investigate the effect of politeness expectations on numerical probabilities as face-management- or likelihood-devices, and also to investigate the effect of these interpretations on risk perception. Politeness concerns have been manipulated directly by using two different descriptions of the personality of the speaker. Results showing an effect of politeness on risk perception based on numbers will allow the application of politeness theory principles to a broader range of formats of risk communication. Contrary results will allow the limits of the theory's application to be defined more clearly and could contribute risk communication guidelines.

2. Experiment 1

This experiment had two objectives: (1) to investigate whether numerical probabilities could be interpreted as a face-management device in a face-threatening communication situation, and (2) to compare risk perception and severity between face-management- and likelihood-communication interpretations. It was expected that politeness concerns would have a positive effect on risk perception and on severity perception.

2.1. Method

2.1.1. Participants

One hundred and ninety three Amazon Mechanical Turk users took part in the experiment. They were Americans aged from 18 to 82 ($M = 33.73$, $SD = 13.18$). Most of the participants were female (59.9%), currently working (68.9%) and Caucasian (84.5%). Education ranged from a General Educational Development degree (GED; 19.8%) to a doctoral degree (2.6%) with most of the participants having some degree of higher education (80.2%).

2.1.2. Design, material and procedure

Hypotheses were tested in a 2 (politeness concerns: tactful speaker vs. blunt speaker) × 2 (format of risk communication: verbal vs. numerical) mixed design. In a within-subjects design, politeness concern was manipulated by the personality of the speaker which was either tactful (i.e., *a polite person, who tends to soften any bad news or criticism*) or blunt (i.e., *a person who tends to give his opinion bluntly, even bad news or criticism*). A similar manipulation of politeness concern was used by Demeure, Bonnefon, and Raufaste (2009) (Study 1) to investigate the effect of politeness concerns on the interpretation of an ambiguous conditional statement (e.g., if the water level is low, then the machine stops automatically) as providing either alternative suggestions or corrections. In a between-subjects design, the format of the probability was either numerical (i.e. *there is a 50% probability*) or verbal (i.e., *it is possible*).

The procedure was adapted from Juanchich et al. (2012). Participants read two scenarios, each followed by a speaker intention measure as a manipulation check, a question measuring subjective probability and finally by a measure of severity, each presented on separate screens. Participants imagined that they had invested their savings in the stock market and were being advised by a tactful [or blunt] friend who was a stock broker (e.g., *it is possible that the stock will lose its value*). Participants then assessed the degree of certainty of the speaker, by providing a minimum and a maximum probability (P_{min} and P_{max}) on two axes ranging from 0 to 100% by increments of 1. The probability question read as follow: *When communicating a likelihood, one often has a range of probabilities in mind. Accordingly, John may have in mind a range of probabilities that the stocks will lose their value. Please state below the lowest and the highest probability John has in mind*. Next, participants assessed the speaker's intention when using the risk quantifier on an axis labeled 0: *The speaker is communicating the likelihood he has in mind*, 100: *The speaker does not want to say harshly that the stock will lose its value*. The interpretation question was "In your opinion, why is your friend saying '[risk quantifier]' that your stocks will lose their value?". Finally, participants rated to what extent the news given was bad on an axis labeled −5: *very bad news* to +5: *very good news*. The scores were then reversed so greater scores reflected a greater severity.

After reading the tactful speaker vignette and responding to the three questions (i.e., probability, interpretation and severity), participants read the blunt speaker scenario and responded to the same three questions. The order of presentation of the two vignettes was counterbalanced.

Finally, participants responded to socio-demographic questions.

Unless stated otherwise, all hypotheses across all experiments have been tested using the conventional level of $\alpha = 0.05$ and 95% confidence intervals.

2.2. Results and discussion

The politeness concern manipulation was successful, as shown by the mean responses on the perceived speaker intention scale, where lower scores reflected likelihood intention and higher scores face-management intention. Participants perceived an increased face-management intention in the polite condition than in the likelihood-communication condition ($M = 69.33$, $SD = 23.95$ and $M = 32.26$, $SD = 28.68$). We conducted a variance analysis for repeated measures to study the effect of the politeness manipulation and of the risk communication format on the perceived intention of the speaker on the speaker intention score. Results showed that politeness affected the perceived intention of the speaker ($F(1, 191) = 208.84$, $p < .001$, $\eta^2_{par} = .52$), whereas format did not significantly affect the perceived intention of the speaker ($F(1, 191) < 1$). Finally, the analysis showed that the effect of politeness concerns did not vary according to the format of communication of risk ($F(1, 191) < 1$).

Fig. 1 depicts the average minimal, maximal and mean probability perceptions (Mean (P_{min}, P_{max})), as a function of the personality of the speaker (blunt vs. polite) and the risk communication format. As shown in Fig. 1 (left panel), on average the minimal, the maximal

and the mean probabilities were higher when the speaker was polite than when the speaker was blunt. Moreover, the severity perception of the news was also greater when the speaker was polite ($M=-2.95$, $SD=1.75$) than when he was blunt ($M=-2.12$, $SD=2.06$). The pattern was consistent for verbal probability (respectively, $M=-2.90$, $SD=1.80$; $M=-1.89$, $SD=2.01$) and numerical probability (respectively, $M=-3.00$, $SD=1.73$; $M=-2.33$, $SD=2.09$).

A two way MANOVA was conducted with politeness concerns and format of risk communication as independent variables, together with five dependent variables: the minimal probability provided, the maximal one, along with the mean probability, the probabilistic range and the perceived news severity. The multivariate tests revealed significant main effects of politeness of the speaker and the format of risk (respectively, $F(3, 189)=17.85$, $p<.001$, $\eta^2_{par}=.22$ and $F(3, 189)=10.15$, $p<.001$, $\eta^2_{par}=.14$). Politeness concerns and the format did not interact ($F(3, 189)=2.16$, $p=.095$, $\eta^2_{par}=.03$). The univariate tests provided more specific results. Politeness concerns had an influence on the maximal probability perception ($F(1, 191)=26.86$, $p<.001$, $\eta^2_{par}=.12$) and on the mean probability perception ($F(1, 191)=21.30$, $p<.001$, $\eta^2_{par}=.10$), but not on the minimum one ($F(1, 191)=2.49$, $p=.116$, $\eta^2_{par}=.01$). Moreover, politeness had an effect on the range covered ($F(1, 191)=12.34$, $p=.001$, $\eta^2_{par}=.06$) and the severity perception ($F(1, 191)=38.72$, $p<.001$, $\eta^2_{par}=.17$). The format used to communicate risk had an effect on the maximal probability perceived ($F(1, 191)=8.43$, $p=.004$, $\eta^2_{par}=.04$) and on the probabilistic range covered by the risk quantifier ($F(1, 191)=25.06$, $p<.001$, $\eta^2_{par}=.12$). The format of communication did not have an effect on the minimal probability, mean perceived probability, or the severity ($Fs<1.23$). Verbal probabilities elicited a greater upper bound of probabilistic range, along with a larger range than numerical probabilities.

In addition, we conducted two within-subject mediation analyses (Judd, Kenny, & McClelland, 2001) to test whether the effect of verbal probability interpretation on severity perception was mediated by the increased risk perception and whether the effect of the interpretation on risk perception was mediated by severity perception.

In previous investigations on the effect of politeness on risk perception and severity perception, it has been suggested that face-management interpretations increased probability perception which in turn increased the perceived severity of the news (Juanchich et al., 2012). Despite evidence that politeness affected both probability and severity perceptions and that the two were correlated, these two mediation hypotheses have not been tested (Juanchich et al., 2012, Experiment 3). According to the first mediation hypothesis, the perception that a speaker has a face-management intention increases risk perception which in turn increases the severity perception of the prediction. For example, a "possible investment loss" prediction given by a tactful speaker constitutes more severe news than when given by a blunt speaker because the tactful speaker is believed to sugar-coat a likely investment loss.

An alternative mediation hypothesis posits that politeness concerns increase severity perception that in turn increases risk perception. It is therefore possible that the use of face-management strategies would be used as a proxy of the situation severity. For example, if Chris appears to be tactful to announce the result of an exam to his student Gill, Gill could infer that she had a bad grade. The effect of severity on risk perception is one of the consequences of the asymmetric loss function (Harris et al., 2009).

To test these two mediation hypotheses in the within-subject design, we followed the procedure outlined by Judd et al. (2001). As recommended by these authors, we conducted a moderation analysis along with the mediation one. First, for the mediation analysis, we computed the difference score for severity and probability perception between the polite and blunt conditions ($\text{DiffSeverity}=\text{Severity}_{\text{polite}}-\text{severity}_{\text{blunt}}$ and $\text{DiffProba}=\text{Probability}_{\text{polite}}-\text{probability}_{\text{blunt}}$). For the moderation analysis, we computed the sum scores for the dependent and mediator variables that were subsequently centered ($\text{SumSeverity}=\text{Severity}_{\text{polite}}+\text{severity}_{\text{blunt}}$ and $\text{SumProba}=\text{Probability}_{\text{polite}}+\text{probability}_{\text{blunt}}$). To assess the effect of politeness on severity and probability, we conducted two one-sample-t-tests against a value of zero on DiffSeverity and DiffProbability. These analyses confirmed that the speaker's politeness had a significant effect on severity ($M=0.82$, $SD=1.86$; $t\ (192)=6.14$, $p<.001$, CI [0.56, 1.09], $d=0.44$) and on probability ($M=4.72$, $SD=14.70$, $t\ (192)=4.46$, $p<.001$, CI [2.63, 6.82], $d=0.32$).

Subsequently, the mediation of the effect of politeness on probability by severity was tested by a regression of DiffProba and ZSumProba on DiffSeverity. ZSumProba was entered in the regression to control for a possible moderation role of the effect of politeness on severity by the perceived probability, while testing for the hypothesized mediation. Results showed that ZSumProba did not affect DiffSeverity, ruling out the possibility that probability played a moderating role ($B=0.03$, $t<1$). Results also showed that probability affected severity perception ($B=0.50$, $t=5.98$, $p<.001$, CI [0.03, 0.07], $d=.43$). Finally, this analysis showed that when controlling for the significant effect of probability on severity, politeness still affected severity perception, as indicated by the intercept value which was significantly different from zero ($B=0.59$, $t=4.52$, $p<.001$, CI [0.33, 0.84], $d=.14$). Overall, the mediation analysis indicated that the effect of the speaker's politeness on the perceived severity of the news was partially mediated by the perceived probability that the stocks would lose their values.

The mediation of the effect of politeness on probability by severity was tested by a regression of DiffSeverity and ZSumSeverity on DiffProba. ZSumSeverity was entered in the regression to control for a possible moderation role of the effect of politeness on probability by the perceived severity, while testing for the hypothesized mediation. Results showed that ZSumSeverity did not affect DiffProba, ruling out the possibility that severity played a moderating role ($B=-0.01$, $t<1$). Results also showed a significant main effect of severity on probability perception ($B=3.13$, $t=5.98$, $p<.001$, CI [2.10, 4.17], $d=.43$). When controlling for the significant effect of severity on probability politeness still affected severity perception, as indicated by the intercept value which was significantly different from zero ($B=2.14$, $t=2.01$, $p=.046$, CI [.0.37, 4.24], $d=.14$). Overall, the mediation analysis indicated that the effect of the speaker's politeness on the perceived probability of investment loss was partially mediated by the perceived severity of the news.

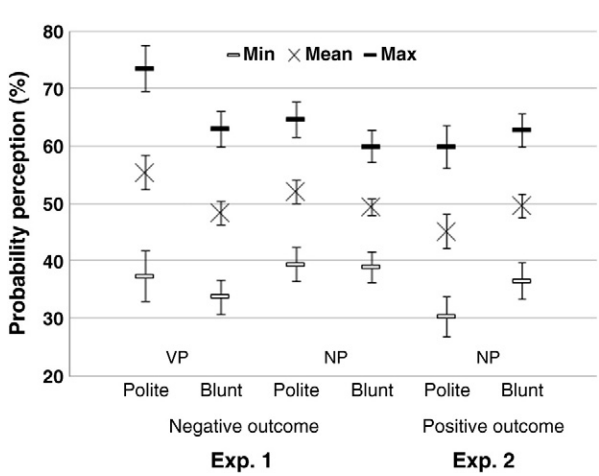

Fig. 1. Effect of politeness expectations on minimal, mean and maximal levels of perceived probability (in mean %) for verbal and numerical negative outcome predictions (Experiment 1) and for a numerical positive outcome prediction (Experiment 2).

Results showed that politeness concerns could stem from the speaker's personality manipulation (blunt vs. tactful) and determine both the verbal and numerical risk quantifier interpretations as likelihood-communication or face-management devices. Tactful speakers were perceived as communicating a greater probability than blunt speakers, independent of the format of risk communication. The average probabilistic range covered by verbal probabilities was larger than for numerical probabilities (37.12 vs. 23.42). This was possibly because of an effect of anchoring numerical value; participants reading a 50% probability may have had less inclination to deviate from this number when assessing the probability communicated by the speaker on a probability scale. The elastic probabilistic meaning of verbal probabilities may also explain why they covered a wider range than numerical ones (Budescu, Weinberg, & Wallsten, 1988; Piercey, 2009; Windschitl & Weber, 1999). Despite the fact that numerical probabilities covered a lower range than the verbal ones, the fact remains that this range was relatively broad (between 21.34 and 25.50). Thus, this highlights the fact that the numerical format may be vaguer than traditionally believed.

Finally, tactful speakers were perceived as communicating worse news than blunt speakers, partly because of the probability inflation created by the face-management interpretation. Device interpretations also directly affected severity perception, possibly because the mere use of the face-management strategy indicated that the situation was so severe that it merited a softening strategy. It is also possible that the face-management interpretation increased the negative outcome accessibility and thus severity perception (Risen & Gilovich, 2008).

3. Experiment 2

This experiment had two main aims. The first aim was to test whether the politeness effect occurs while controlling for the individual differences in risk perception. Specifically, numeracy (i.e. the ability to understand and work with numbers) and cognitive reflection ability (i.e., the ability to resist reporting the response that first comes to mind) are often the two major individual difference variables considered in research investigating the determinants of numerical probability perception or probability updating (e.g., Liberali, Reyna, Furlan, Stein, & Pardo, 2011; Peters et al., 2006; Reyna, Nelson, Han, & Dieckmann, 2009; Sirota & Juanchich, 2011). For the former, it has been shown that people with low numeracy tend to overestimate the risk of negative health related events (Reyna et al., 2009) and are sensitive to contextual effects in probability estimation (Peters et al., 2006). For the latter, it has been demonstrated that people with low cognitive reflection are sensitive to the contextual effects such as framing effect and are generally more prone to biases in the decision-making domain (Toplak, West, & Stanovich, 2011). Therefore, it could be argued that the effect observed in Experiment 1 has occurred because of relatively low numeracy/cognitive reflection level in our sample. To distinguish variation in risk estimates caused by a lack of numeracy or cognitive reflection from a variation caused by politeness expectations, this experiment featured a measure of these two abilities.

The second aim of this experiment was to test whether the effect of politeness expectations on the interpretation of numerical probabilities and associated risk perception is robust. Specifically, the experiment was tailored to replicate the finding of Experiment 1 in a between-subject design and to extend it to a different sample (Slovak students) and a positive outcome (i.e., there is a 50% probability that *your car will be ok*).[1] Positive outcome predictions can be quite threatening, as presenting a desired outcome that is qualified as uncertain (e.g., it is not certain that people enjoyed your presentation; Juanchich et al., 2012). Positive outcome predictions can thus be perceived as negatively valenced predictions that threaten the positive face of the recipient. Speakers uttering positive outcome predictions are thus expected to use face-management strategies and recipients are then likely to opt for face-management interpretations. In positive outcome predictions, the numerical probability will still be interpreted according to the personality of the speaker (i.e., polite vs. blunt), albeit the effect of the face-management interpretations is expected to differ from negative outcome predictions. Indeed, tactful speakers announcing an uncertain good outcome may tend to communicate a greater probability than the one they have in mind (e.g., "it is possible" instead of "there is a chance"). Consequently, when a speaker is perceived to tactfully give the probability of a good outcome, hearers may adjust their probability perception downward to match that of the speaker (Juanchich et al., 2012).

3.1. Method

3.1.1. Participants

Participants were 212 Slovak undergraduate students in Economics (126 females) aged 18 to 28 years ($M = 21.32$, $SD = 1.14$). Slovak students were found to be similarly sensitive to politeness needs as British individuals (Juanchich et al., 2012).

3.1.2. Design, material and procedure

Participants completed the web-based questionnaire at the end of a class. In a between-subjects design, participants read a scenario involving either a tactful or a blunt speaker giving bad news in the form of an uncertain good outcome. The manipulation of politeness concerns by the personality of the speaker was the same as in Experiment 1. Participants imagined that they had bought a used car without warranty and were advised by a friend who was a car mechanic: "There is a 50% chance that your car will be ok". Participants assessed the degree of certainty of the speaker and provided their perception of the speaker's intention when using the numerical probability (likelihood-communication vs. face-management) as in Experiment 1. Then, participants responded to the 11 items of the Lipkus Numeracy Scale (Cronbach's α between .70 and .75; Lipkus, Samsa, & Rimer, 2001) and to the three items of the Cognitive Reflection Test (Frederick, 2005), both adapted in Slovak by Sirota and Juanchich (2011). The reliability of the scales was satisfactory with Cronbach's $\alpha = .85$ for numeracy and Cronbach's $\alpha = .62$ for cognitive reflection. The scores of numeracy ranged from 1 to 11 with $M = 8.57$, $SD = 2.07$, and $Mdn = 9$, and scores of cognitive reflection ranged from 0 to 3 with $M = 0.89$, $SD = 1.02$, and $Mdn = 1$. Thus, the reliability and the variability of these measures were similar to those reported elsewhere with recognized impact on risk perception and updating (Liberali et al., 2011; Sirota & Juanchich, 2011).

3.2. Results and discussion

The manipulation of politeness concerns was effective according to the checking item which measured participants' perceptions of speakers' intentions on a scale ranging from likelihood-communication to face-management. Participants scored higher in the polite condition ($M = 66.06$, $SD = 25.01$) than in the blunt condition ($M = 57.42$, $SD = 27.05$; t (204) $= 2.37$, $p = .019$, $d = 0.33$, CI [1.45, 15.84]). It is interesting to note that, overall, participants considered a person to be quite tactful when describing the probable occurrence of a positive outcome (i.e., the car being ok) whereas, in study 1, participants were less inclined to attribute face-management intention to a speaker describing a negative outcome (i.e., losing an investment). Although the different nature of the outcomes might be responsible for the different device interpretations, it is also possible that framing a prediction in positive terms (e.g., there is a 50% chance that your car will be ok) is less threatening

[1] To test whether the outcome "the car being ok" as tested in our vignette was indeed perceived as a positive outcome, a sample of 33 young adults from Slovakia assessed the extent to which the outcome was positive/negative on a 7-points Likert scale (1 = positive, 7 = negative). The outcome "the car is ok" as used in our car vignette was perceived by 29 out of 33 participants as a positive outcome (M = 2.1, Mo = 1, SD = 1.2, CI [1.7, 2.5]). To have a baseline of comparison (i.e., to be sure that the assessment of our participants is not biased to one side of the scale) we asked the same participants to assess the outcome from Exp. 1. The outcome "the stock has lost its value" as used in the stock vignette was perceived by 31 out of 33 participants as a negative outcome (M = 5.9, Mo = 6, SD = 0.9, CI [5.6, 6.3]).

than framing the prediction in negative terms (e.g., there is a 50% chance that your car will break down). Findings of Juanchich et al. (2012) suggested a similar pattern of results.

In line with our hypothesis, the politeness manipulation affected participants' probability perceptions in the predicted direction. Participants perceived a lower mean probability that the car would be ok when the speaker was tactful ($M = 45.50$, $SD = 15.40$) than when the speaker was blunt ($M = 49.35$, $SD = 11.57$). The effect of politeness on minimal, mean and maximal perceived probabilities is depicted in Fig. 1 (right panel). To test the hypothesized effect of politeness, a one-way MANOVA was conducted with politeness as the independent variable and the min, max, range and mean probability perceptions as dependent variables. The multivariate analyses showed a global effect of politeness on the dependent variables ($F(2, 209) = 3.51$, $p = .032$, $\eta^2_{par} = .03$). The tests of univariate analysis of between-subjects effects showed more specifically that the politeness of the speaker had an effect on the minimal probability perception and on the average probability perception (respectively, $F(1, 210) = 6.53$, $p = .011$, $\eta^2_{par} = .03$; $F(1, 210) = 5.91$, $p = .016$, $\eta^2_{par} = .03$) but not on the maximal probability, nor on the probabilistic range (respectively, $F(1, 210) = 1.40$, $p = .238$, $\eta^2_{par} = .01$ and $F(1, 210) = 1.18$, $p = .278$, $\eta^2_{par} = .01$). Politeness expectations seem to stretch only one of the bounds of the perceived probabilistic range. The bound that is adjusted seems to depend on the valence of the outcome predicted: the lower bound is adjusted lower for a positive outcome and the higher bound is adjusted upwards for a negative outcome.

To control for the effect of numeracy and cognitive reflection, we conducted a one-way MANCOVA integrating numeracy and cognitive reflection as independent covariates in addition to the variables integrated in the variance analysis conducted previously. The multivariate analysis confirmed the effect of politeness ($F(2, 189) = 3.38$, $p = .036$, $\eta^2_{par} = .03$). Moreover, results did not show a significant effect of numeracy or cognitive reflection on probability perception ($F < 1$ and $F(1, 193) = 1.48$, $p = .230$, $\eta^2_{par} = .01$, respectively). These results are interesting in several ways. First of all, the results demonstrate robustness of the studied effect—the effect of politeness on risk perception was statistically significant even when controlling for numeracy and cognitive reflection ability. In addition to that, these findings help to reveal the nature of the politeness effect. Specifically, in contrast to some other effects on risk perception (e.g., Peters et al., 2006), the lack of the effect of numeracy points to the fact that the politeness effect is independent of the ability to understand or to work with numbers. Thus, the nature of overestimation is not driven by any lack of numerical knowledge or skill. Moreover, the lack of the effect of cognitive reflection shows that the politeness effect is not that type of miserly-processed error as described by heuristics-and-biases literature (Toplak et al., 2011). On the contrary, the effect represents adjustment ruled by the pragmatics of human communication, which enables matching with the probability the speaker bears in mind.

Results demonstrated that a polite speaker describing the numerical probability of the occurrence of a positive outcome (i.e., buying a second hand car that will be ok) is perceived as communicating a lower probability than a blunt speaker. The probability that is adjusted based on the polite personality of the speaker is the minimal probability of the perceived probabilistic range. It can be considered that the outcome "buying a used car that will be ok", is not an extremely positive outcome (neither extremely valued, nor desired). Predictions focusing on outcomes that are more positive, such as "there is a 50% chance that you will have a healthy baby" are expected to elicit a similar adjustment direction of risk perception (i.e., the hearer will adjust the chance of the baby being healthy downwards, believing that the doctor is exaggerating this chance to be tactful). Yet, because this sentence may be perceived as more threatening than a less extreme positive outcome prediction, it could trigger a greater frequency of face-management interpretations and these interpretations may have a greater effect on risk perception.

Results of Experiments 1 and 2 consistently showed that (1) participants interpreted the numerical probability 50% as a face-management device when participants had high politeness expectations and that (2) face-management interpretations of a 50% chance led to an adjusted risk perception compared to likelihood-communication ones which did not. The present results indicate that differences in numeracy or cognitive reflection did not account for subjective probability deviation from the given probability.

4. Experiment 3

So far only one numerical probability has been investigated: a 50% chance. This next experiment aimed to test whether politeness expectations also affect numerical probabilities other than 50%. First, it is arguable that the effect of politeness occurs only for middle range expressions, which fall in the 50% "blip", and amount to "I don't know" (Beach & Metzger, 1997; Fischhoff & Bruine de Bruin, 1999). Second, these expressions are certainly the closest to verbal probability, as one can easily say "there is a 50/50 chance" without really intending a numerical meaning (Dhami & Wallsten, 2005). In addition, the present study took the pragmatic perspective a step further by manipulating the need for politeness based on the needs of the two conversational partners. This manipulation aimed to stress a dialogical form of communication, where two interlocutors interact to build the meaning of the discussion. The rationale for stressing the dialogical nature of communication was inspired by psycholinguistics research, where dialogs have been explored as an alternative model for language understanding (Clark, 1996; Garrod & Pickering, 2004).

4.1. Method

4.1.1. Participants

One hundred and thirty five Amazon Mechanical Turk users took part in the experiment. They were Americans aged from 18 to 82 ($M = 33.73$, $SD = 13.18$). Most of the participants were females (56.1%), currently working (68.2%) and Caucasian (80%). Education ranged from a GED (19.8%) to a doctoral degree (2.6%) with most of the participants having some degree of higher education (81.2%). Four participants did not report their socio-demographics.

4.1.2. Design, material and procedure

Participants read all three vignettes presented in a randomized order (vignettes: car, job application and investment). Each vignette described a person giving bad news to a friend. The bad news was either a 30% chance that a new car would break down, a 50% chance of being rejected in a job application or a 70% chance of losing on an investment (see vignettes in Appendix). In the between-subjects design, the conversational context stressed either a need for politeness or a need for plain speaking. The conversational context was manipulated by the conversational expectations of the interlocutors, combining hearers' needs and speakers' intentions. The speaker either perceived that the hearer (i.e., participant) could be easily upset and was thus eager to soften his prediction or perceived that the hearer expected clear information and thus tried to speak plainly. Participants read either "Your friend thinks that you want the plain information about your car and thus tries to plain speak when he says" in the likelihood communication condition or "Your friend thinks that you may get easily upset and thus tries to soften his prediction" in the face-management condition. This manipulation of politeness concerns was previously used in Juanchich et al. (2012, Study 4).

For each vignette, participants judged the degree of certainty of the speaker on a probability scale ranging from 0 to 100% by increments of 1. The question was "In your opinion, what is the probability that your car will break down soon according to your friend?".

Finally, participants reported their socio-demographic characteristics.

4.2. Results and discussion

Fig. 2 shows the mean risk perception for 30%, 50% and 70% negative outcome predictions as a function of whether the conversational context stressed the need for politeness or the need for plain speaking. In the three contexts and irrespective of the probability of the outcome, participants perceived that the speaker communicated a greater probability when the interlocutors needed politeness than when they needed plain speaking.

As described in Fig. 2, participants perceived greater risk across the three vignettes when the context stressed the need for face concerns than when the context stressed the need for plain speaking. Participants perceived a higher risk of having their car break down (+5.87%), not being accepted for a job (+8.22%) and their stock losing its value (+7.06%). These differences, were statistically significant according to three t-tests (respectively, t (135.15) = 3.02, p = .003, *CI* [2.03, 9.71], d = 0.51; t (119.48) = 4.59, p < .001, *CI* [4.66, 11.78], d = 0.80; t (135) = 3.34 p = .001, *CI* [2.88, 11.24], d = 0.57).

Participants provided subjective estimates based on three probabilities: 30%, 50% and 70%. It is interesting to note that in the likelihood-communication condition, the estimates of participants were quite accurate or slightly below the given probability for the three probability magnitudes (i.e., 30.73%, 48.06%, and 67.34%) whereas in the politeness condition, the estimates were slightly higher than the given probability (i.e., 36.60%, 56.28%, and 74.40%).

Results showed that the conversational context had a significant effect on risk estimates in the three contexts. The fact that this study did not feature the assessment of the speaker's intention prior to the probability assessment supports the view that it is indeed the politeness manipulation that affects risk perception and not the effect of the speaker intention checking item.

The present research provides further evidence that numerical probabilities can be perceived as communicating an inaccurately higher likelihood in conversational contexts which stress the need for politeness.

5. Experiment 4

Our previous findings focused on numerical probabilities expressed as percentages, showing that low, medium and high numerical probabilities are sensitive to politeness expectations, in negative and positive outcome predictions. The following study aims to investigate whether the effect of politeness expectations can be broadened to other presentational formats of numerical probabilities such as chance ratios (e.g., one chance in three).

Moreover, in this study we reversed the order of the probability and interpretation questions. In Experiments 1 and 2, the device question was placed after the probability question, giving the possibility that participants adapted their interpretations of the speaker's intention as a function of the subjective probability they reported. The present study may rule out the possibility that the politeness effect could result from demand characteristics.

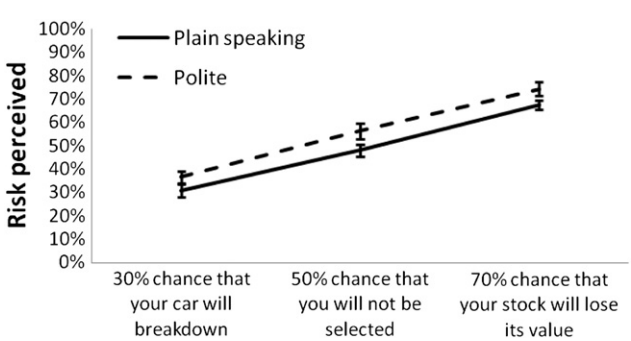

Fig. 2. Mean risk perception for 30%, 50% and 70% negative outcome predictions as a function of the need for politeness.

5.1. Method

5.1.1. Participants

Eighty three Amazon Mechanical Turk users took part in the experiment. They were Americans aged 18 to 65 (M = 35.26, SD = 12.34). Most of the participants were females (59.8%), currently working (69.5%) and White/Caucasian (79.3%). Education ranged from below high school (1.2%) to a Master's degree (4.9%); most of the participants had at least some degree of higher education (78.1%).

5.1.2. Design, material and procedure

Participants read only one scenario, featuring a one in three chance ratio in a polite or a plain speaking condition. The bank investment context from Experiment 1 was used here. The need for politeness was manipulated as in Experiment 3 by means of the conversational context in a between subjects design. Participants assessed the speaker's intention by selecting one of two intentions: *the speaker wants to give his degree of certainty*, and *the speaker wants to communicate tactfully the probability that your stock will lose its value*. Then, participants assessed the probability that the stock would lose its value by placing a cursor on an axis ranging from 0 to 100% per increments of 1. The question was phrased as in Experiment 3 (i.e., *In your opinion, what is the probability that your stock will lose its value*?).

5.2. Results and discussion

The numerical ratios of chances were perceived as a face-management strategy more often in the context stressing politeness needs (71.4%) than in the context stressing the need for plain speaking (59.0%). Yet, in both conditions, most of the participants considered that the speaker was saying "there is a 1 chance in 3 that your stock will lose its value" for a face-management purpose (to tactfully state a probability). The effect of the conversational context was marginally statistically significant (χ^2 (1, N = 167) = 2.83, p = .093, φ = 0.13). These results contrast findings in previous experiments in which the proportion of face management was on average lower, especially in the plain speaking context.

In the conversational context stressing politeness expectation, participants perceived that the speaker was communicating a greater probability (M = 38.59, SD = 12.32) than did participants in the context stressing the need for likelihood-communication (M = 33.34, SD = 9.74; t (75.95) = 2.14, p = .036, d = 0.43).

These results therefore further corroborate the finding that politeness expectation determines risk perception, even with precise numerical ratios. The results replicated the effect of politeness in a between-subjects design where participants judged first the speaker's intention and then the probability question, ruling out that the effects previously observed were due to the order of the tasks. Moreover, the present results support that politeness expectations are at play when one is interpreting numerical probabilities in general (e.g., percentage, ratios).

6. General discussion

Politeness theory predicts the use of hedging strategies to give bad news. In the risk communication context, this theory and the research stemming from it have previously focused only on the use of linguistic expressions (Bonnefon, Feeney, & Villejoubert, 2009; Bonnefon & Villejoubert, 2006; Juanchich et al., 2012). The present research aimed to test whether numerical formats of risk communication can also be interpreted as face-management strategies and lead to a risk perception that contrasts likelihood-communication interpretations.

Results from four experiments support the view that the pragmatic face-management function can be performed by numerical probabilities as well as by the more traditionally studied verbal probabilities (e.g., it is possible). This phenomenon is applied for low, middle range and high probabilities and also for different numerical formats such as percentage or chance ratio. Moreover, the politeness expectations have implications in different populations (American and Slovak) and go beyond numeracy or cognitive reflection abilities. The need for politeness was manipulated either by the personality of the speaker in Experiments 1 and 2 (i.e., tactful vs. blunt) or by the needs of the two interlocutors in Experiments 3 and 4. Results broadened the range of determinants shaping politeness expectations. They also supported the conclusion that even numerical probabilities are under the spell of politeness expectation, which holds implications for risk communication policies.

6.1. Theoretical implications for politeness and risk perception

The manipulation of politeness expectations by the personality of the speaker and the expectations of the conversational partners widened the body of determinants of politeness expectation. So far the determinants of politeness expectation have been mostly limited to three contextual factors suggested by the politeness theory (i.e., social distance, degree of imposition and power of the hearer over the speaker; Brown & Levinson, 1987).

Moreover, the politeness expectation manipulation has helped clarify the fact that the effect of politeness is different from the effect of other contextual variables such as base rate or severity. In previous methodologies investigating the effect of politeness on risk perception, the need for politeness was manipulated by the severity of the outcome. These methods were based on the assumption that severe bad news such as a prognosis of deafness would trigger more face-management interpretations than less severe bad news such as insomnia (Bonnefon & Villejoubert, 2006, Juanchich et al., 2012). This manipulation could have been considered as a methodological limitation, as the severity of an outcome may also directly contribute to the risk increase in the severe condition (Harris & Corner, 2011; Harris et al., 2009; Vosgerau, 2010; Weber & Hilton, 1990). Moreover, outcome severity is often confounded with the base rate of occurrence of the outcome, which has also been found to affect risk perception (Windschitl & Weber, 1999). The manipulations of politeness expectations based on the personality or on the conversational context have provided reliable evidence of the effect of politeness expectations beyond the outcome's severity and its base rate.

The effect of politeness concerns on risk perception has been demonstrated using a numerical probability scale. This result contrasts previous findings, showing that contextual effects on risk perception (e.g., base rate) disappear when measured on a numerical probability scale rather than a verbal probability scale (Windschitl & Wells, 1996). It has been assumed that numerical probability scales measure a rule based degree of certainty that is less sensitive to contextual factors than verbal probability scales which are seen to measure a more intuitive degree of certainty (Windschitl & Weber, 1999; Windschitl & Wells, 1996). The fact that verbally labeled scales are expected to elicit a more intuition based degree of certainty than numerical scales suggests that using a verbal probability scale could unveil a stronger effect of politeness than the one observed here with a numerical probability scale.

6.2. Applied implication of the perceived functions of numerical probabilities

The present results show that the interpretation of numerical probabilities as face-management or likelihood-communication devices determines risk perception. A tactful speaker announcing that participants' stocks had a 50% probability of losing their value was deemed to communicate a greater risk than the same announcement being expressed by a blunt speaker. This discrepancy in risk and severity perception may have important consequences in terms of decision-making, especially knowing that numerical formats of risk communication give an illusion of precision.

It is worth noting that there was a greater agreement between interpretations of numerical probabilities when the politeness expectation was high than when the need was for plain speaking. On average, almost 80% of participants agreed that the speaker intended to soften bad news when politeness expectations were raised, whereas only 60% of participants agreed that a more direct speaker aimed to simply communicate likelihood. These results are in opposition to those observed with verbal probabilities, where the rate of agreement was lower in contexts with a high need for politeness, manipulated by the severity of the predicted outcome (Bonnefon et al., 2011). It is possible that the difference is caused by the format (verbal vs. numerical) or by the different manipulations of politeness expectation which may trigger more or less agreement about the intention of the speaker. Yet, the greater consistency of interpretations of numerical probabilities given in contexts stressing politeness expectation indicates that hearers may have greater problems in understanding the intention of a typically plain speaker using numerical probabilities.

The use of numerical probabilities has been recommended to avoid polite risk miscommunication (e.g., Bonnefon & Villejoubert, 2006). Previous evidence indeed points in this direction. Risk perception accuracy was for example substantially improved when verbal probabilities were accompanied with numerical values compared to verbal probabilities only in different climate change contexts (Budescu, Broomell, & Por, 2009; Budescu, Por, & Broomell, 2012). However, the present results show that the risk of polite misunderstandings cannot be completely eliminated by using numerical rather than verbal probabilities. Therefore, formats other than numerical or verbal might also contribute to avoiding confusion between two possible interpretations. For example, figures might be more appropriate to elicit an accurate risk perception and decrease risk miscommunication (Galesic, Garcia-Retamero, & Gigerenzer, 2009; Lipkus & Hollands, 1999, see also Sedlmeier & Hilton, 2011). Indeed, figures such as bar charts might not be ruled by the same polite conversational expectations as numbers and words.

6.3. Concluding comment

From a purely normative standpoint, the probabilistic adjustment between the face-management and the likelihood-communication interpretations could be considered as a bias, representing a violation of the normative invariance principle. However, this adjustment might also be considered as an ecologically valid pragmatic adjustment enabling a match with the probability the speaker has in mind. Further investigation might focus on strategies used by people communicating risk, to question the relevance of the probability adjustment made by the hearer. The combination of findings from the standpoint of both conversational partners could form the basis of more informed recommendations tailored to improve risk communication.

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Appendix

Car vignette

Imagine that you have bought a used car without a warranty. You are not a car mechanic but you think that you have made a good deal. You are showing your new car to a friend who is a car mechanic.

Stock vignette

Imagine that you have invested in the stock market all the saving that you have accumulated over time. You are a novice in such investments but you think that the stocks you bought are very good. You talk about your investment with a friend who is a stock broker.

New job vignette

Imagine that you are applying for the position as manager. You have been unemployed for 6 months and you have high hopes in this application. You are discussing your chances with a friend who is already working as a manager in the company.

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